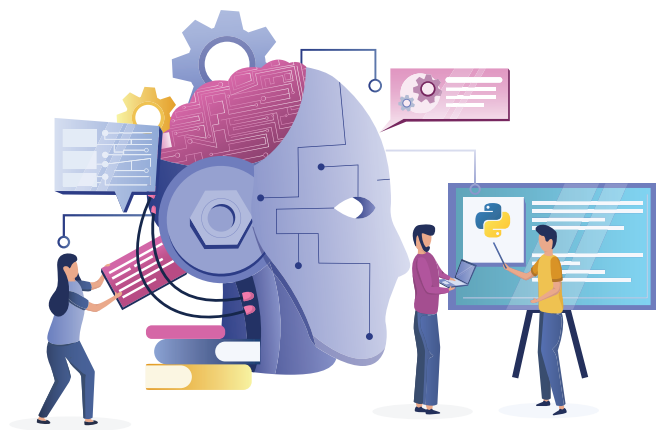


AI: Classical Techniques and Hidden Python Gems

This eleventh article in the AI series begins by providing a brief overview of semi-supervised learning and reinforcement learning. While the series has primarily focused on machine learning based techniques for AI, it is important to discuss classical approaches that do not rely on machine learning to provide a comprehensive understanding of AI. Therefore, this article will explore some of these classical techniques. Later, it will introduce some powerful yet relatively unknown Python libraries and packages that can be used to develop AI based applications.



Besides the commonly used paradigms of supervised and unsupervised learning, there are two other less familiar paradigms in machine learning — semi-supervised learning and reinforcement learning.

First, let us try to understand how semi-supervised learning works. The semi-supervised learning technique lies between supervised learning, which relies solely on labelled data for training, and unsupervised learning, which operates without any labelled data. Semi-supervised learning uses both labelled and unlabelled data to train a model. One of the techniques called label propagation use the known labels to estimate the labels of the unlabelled data points based on their similarity to the labelled data points. In label propagation, the similarity between data points is measured using a similarity function such as the Gaussian

kernel. Recall that we have discussed Gaussian kernels in the previous article in this series. Label spreading is a semi-supervised learning algorithm that is similar to label propagation. Python libraries like Keras, TensorFlow, PyTorch, and scikit-learn are all used to implement models based on semi-supervised learning.

Now, let us explore the working of reinforcement learning (RL). While RL is not as well-known as other machine learning approaches we have discussed, its popularity has recently skyrocketed. This is largely due to the success of ChatGPT, a widely-discussed natural language processing model that employed RL, in addition to supervised learning, for fine-tuning. RL is a sub-field of machine learning that focuses on teaching an agent to take actions that maximise its total reward over time within a

given environment. RL algorithms are utilised to address decision-making problems that are sequential in nature, where an agent interacts with an environment, takes an action, and receives a reward based on the action taken. The ultimate goal for the agent is to learn a policy that will provide the maximum expected cumulative reward over a sequence of actions. RL problems can be represented as Markov Decision Processes (MDP), which provide a mathematical framework for modelling decision-making problems. RL has been effectively applied to a wide range of problems, including game playing, robotics, autonomous vehicles, and recommendation systems. Some popular RL algorithms include Q-Learning, SARSA, etc.

Although Python libraries such as TensorFlow and Keras can be utilised for developing RL based models, we will focus solely on the features of a relatively unknown standard API called 'gym'. Gym was originally created by OpenAI, the same company responsible for the development of ChatGPT. Additionally, a fork of gym, called 'gymnasium', also exists and is actively developed nowadays. The gym/gymnasium interface is simple and capable of representing general RL problems. The Python script shown below (line numbers are included for ease of explanation), illustrating the